

Cooper Pair Phonon Coupling

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PAPER_951: Cooper Pair Phonon Coupling

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** `bcs_{superconductivity_uqff}.py` (CooperPairPhononCoupling) **Calculator:** CooperPairPhononCouplingCalc (CP4 #535) **CVW:** v2.0.0 compliant

Abstract

We derive the effective Cooper-pair coupling strength via SCm phonon exchange at 1.25 THz. The coupling $V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \Phi_{1.25 \text{ THz}}(\omega, \Gamma)$ is modulated by the phonon resonance profile $\Phi = \exp(-(\omega - \omega_{\text{extSCm}})^2 / (2\Gamma^2)) \cdot S_{26}$, yielding pair binding energy $E_{\text{pair}} = 2\Delta(T)$.

1. Coupling Strength

$$V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \exp\left(-\frac{(\omega - \omega_{\text{extSCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

2. Pair Binding Energy

$$E_{\text{pair}} = 2\Delta(T)$$

At on-resonance ($\omega = \omega_{\text{extSCm}}$), $\Phi = S_{26}$ and coupling is maximized.

3. Source Data

- **File:** `bcs_{superconductivity_uqff}.py`
- **Session:** 214
- **CP4 Class:** CooperPairPhononCouplingCalc (#535)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
 2. Cooper, L.N. (1956) — Phys. Rev. 104, 1189
 3. PAPER_949 — BCS Gap Equation
 4. PAPER_950 — BCS Critical Temperature
 5. PAPER_955 — BCS Phonon Resonance
 6. PAPER_957 — Cooper Pair Lagrangian
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Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation driven by this coupling
PAPER_950	T_c depends on V_{eff}
PAPER_952	Spectral ladder channels for pairing
PAPER_955	Resonance Q-factor at ω_{extSCm}

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_{Bi_i}\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	—	Magnetar spin-down
$[SSq]$	0.57	—	BH dynamics
β_i	0.603	—	Multi-system
$\omega_{t\text{extSCm}}$	$2\pi \times 1.25 \text{ THz}$	—	Phonon resonance
F_{UBi}/F_U	0.6	—	Buoyancy-gravity ratio

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Cooper pair binding	$E_{\text{pair}} = 2\Delta$ via $\Phi_{1.25\text{THz}}$	Mapped
Phonon mediation	$V_{\text{eff}} = V_{\text{SCm}} \cdot \Phi(\omega, \Gamma)$	Novel

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Cooper Pair Binding (BCS-SCm Phonon Mediation)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{pair}} = \bar{\psi}(i\partial\!\!\!/ - m)\psi + V_{\text{SCm}}(\bar{\psi}\psi)^2 \cdot \Phi_{1.25\text{THz}}$$

§A.3 Euler-Lagrange Equation of Motion

$$(i\partial\!\!\!/ - m)\psi = -2V_{\text{SCm}}\Phi \cdot (\bar{\psi}\psi)\psi$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ $\Phi_{1.25\text{THz}}$ phonon $\rightarrow$ Cooper pair binding $\rightarrow$ BCS condensate

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Cooper pair density scales with $\rho_t extSCm \cdot S_{26}$.

§B.2 DVP

Prime $p = 2$ encodes pair symmetry.

§B.3 BSH

$$\text{BSH}_{\text{pair}} = \tanh(\beta_i \cdot \Delta/E_0) \cdot S_{26}$$

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold

Paper	Title
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy

17 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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